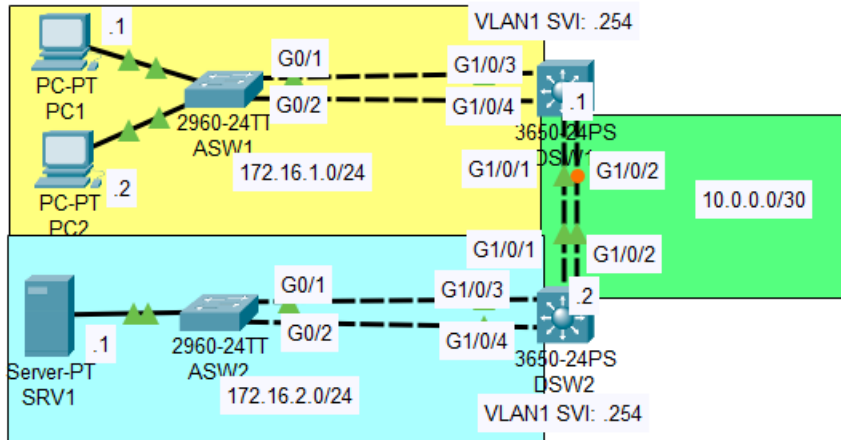




Topology

1. Configure Layer 2 EtherChannel between ASW1 and DSW1 using LACP. Configure it as a trunk.

```
ASW1>en
ASW1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    20481
             Address     0007.EC07.1D30
             Cost        8
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0005.5E1E.5B9D
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Gi0/1	Root	FWD	4	128.25	P2p
Gi0/2	Altn	BLK	4	128.26	P2p
Fa0/1	Desg	FWD	19	128.1	P2p

```
ASW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ASW1(config)#int range g0/1-2
ASW1(config-if-range)#channel-group 1 mode active
ASW1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

ASW1(config-if-range)#int pol
ASW1(config-if)#switchport mode trunk
```

```
ASW1(config-if)#do sh running-config
Building configuration...
```

```
Current configuration : 1231 bytes
```

```
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
```

```
hostname ASW1
```

```
!
!
!
!
!
!
```

```
spanning-tree mode pvst
spanning-tree extend system-id
!
```

```
interface Port-channel1
 switchport mode trunk
interface GigabitEthernet0/1
 switchport mode trunk
 channel-group 1 mode active
!
```

```
interface GigabitEthernet0/2
 switchport mode trunk
 channel-group 1 mode active
,
```

```
ASW1(config-if)#do sh etherchannel summary
```

```
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 1
```

```
Group Port-channel Protocol Ports
```

```
-----+-----+-----+-----
```

```
1      Pol(SD)          LACP  Gig0/1(I) Gig0/2(I)
```

```

DSW1>en
DSW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
DSW1(config)#int range g1/0/3-4
DSW1(config-if-range)#channel-group 1 mode active
DSW1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

DSW1(config-if-range)#int pol
DSW1(config-if)#switchport mode trunk
DSW1(config-if)#do sh eth summ
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol(SU)          LACP       Gig1/0/3(P) Gig1/0/4(P)

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol(SU)          LACP       Gig1/0/3(P) Gig1/0/4(P)
DSW1(config-if)#do show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Pol        on        802.1q         trunking    1

Port      Vlans allowed on trunk
Pol        1-1005

Port      Vlans allowed and active in management domain
Pol        1

Port      Vlans in spanning tree forwarding state and not pruned
Pol        1

```

```

ASW1(config-if)#do sh span
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    20481
            Address     0007.EC07.1D30
            Cost        7
            Port        27 (Port-channel1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
            Address     0005.5E1E.5B9D
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Pol	Root	FWD	3	128.27	Shr
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/1	Desg	FWD	19	128.1	P2p

2. Configure Layer 2 EtherChannel between ASW2 and DSW2 using PAgP. Configure it as a trunk.

```

ASW2>en
ASW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#int range g0/1-2
ASW2(config-if-range)#channel-group 1 mode desirable
ASW2(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

ASW2(config-if-range)#int pol
ASW2(config-if)#switchport mode trunk

```

```

DSW2>en
DSW2#con
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up
f t
Enter configuration commands, one per line. End with CNTL/Z.
DSW2(config)#int range g1/0/3-4
DSW2(config-if-range)#channel-group 1 mode desirable
DSW2(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

DSW2(config-if-range)#int po1
DSW2(config-if)#switchport mode trunk

```

3. Configure Layer 3 EtherChannel between DSW1 and DSW2 using static EtherChannel.

```

DSW1(config-if)#int range g1/0/1-2
DSW1(config-if-range)#no switchport
DSW1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

DSW1(config-if-range)#channel-group 2 mode on
DSW1(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

DSW1(config)#int po2
DSW1(config-if)#ip address 10.0.0.1 255.255.255.252

```

```

DSW2(config-if)#int range g1/0/1-2
DSW2(config-if-range)#no switchport
DSW2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

DSW2(config-if-range)#channel-group 2 mode on
DSW2(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

DSW2(config-if-range)#int po2
DSW2(config-if)#ip address 10.0.0.2 255.255.255.252

```

4. Configure routes to allow the PCs to reach SRV1.

```

DSW1(config-if)#do sh ip route
Default gateway is not set

```

Host	Gateway	Last Use	Total Uses	Interface
ICMP redirect cache is empty				

```

DSW1(config-if)#exit
DSW1(config)#ip routing
DSW1(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Port-channel2
      172.16.0.0/24 is subnetted, 1 subnets
C       172.16.1.0 is directly connected, Vlan1

```

```

DSW1(config)#ip route 172.16.2.0 255.255.255.0 10.0.0.2
DSW1(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Port-channel2
      172.16.0.0/24 is subnetted, 2 subnets
C       172.16.1.0 is directly connected, Vlan1
S       172.16.2.0 [1/0] via 10.0.0.2

```

```
DSW2(config)#ip routing
DSW2(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

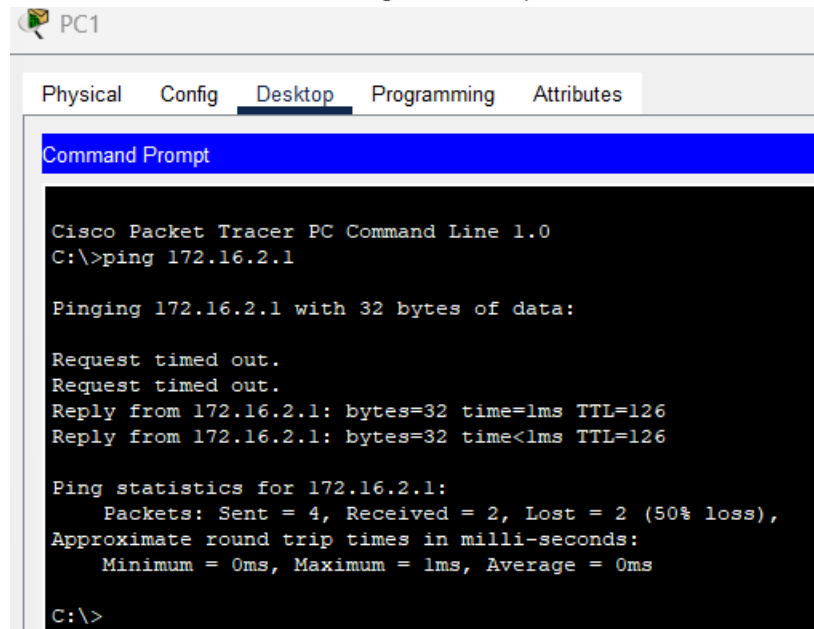
Gateway of last resort is not set

```
      10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Port-channel2
      172.16.0.0/24 is subnetted, 1 subnets
C       172.16.2.0 is directly connected, Vlan1
```

```
DSW2(config)#ip route 172.16.1.0 255.255.255.0 10.0.0.1
DSW2(config)#do sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
      10.0.0.0/30 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Port-channel2
      172.16.0.0/24 is subnetted, 2 subnets
S       172.16.1.0 [1/0] via 10.0.0.1
C       172.16.2.0 is directly connected, Vlan1
```



5. What is the default EtherChannel load-balancing method used on each switch?

```
ASW1#sh etherchannel load-balance
EtherChannel Load-Balancing Operational State (src-mac):
Non-IP: Source MAC address
IPv4: Source MAC address
IPv6: Source MAC address
```

```

DSW1(config)#do sh etherchannel load-balance
EtherChannel Load-Balancing Configuration:
    src-mac

EtherChannel Load-Balancing Addresses Used Per-Protocol:
Non-IP: Source MAC address
IPv4: Source MAC address
IPv6: Source MAC address

```

6. Configure the switches to load-balance based on source and destination IP addresses.

```

ASW1(config)#port-channel load-balance ?
dst-ip      Dst IP Addr
dst-mac     Dst Mac Addr
src-dst-ip  Src XOR Dst IP Addr
src-dst-mac Src XOR Dst Mac Addr
src-ip      Src IP Addr
src-mac     Src Mac Addr
ASW1(config)#port-channel load-balance src-st-ip
^
% Invalid input detected at '^' marker.

ASW1(config)#port-channel load-balance src-dst-ip
DSW1(config)#port-channel load-balance src-dst-ip

```